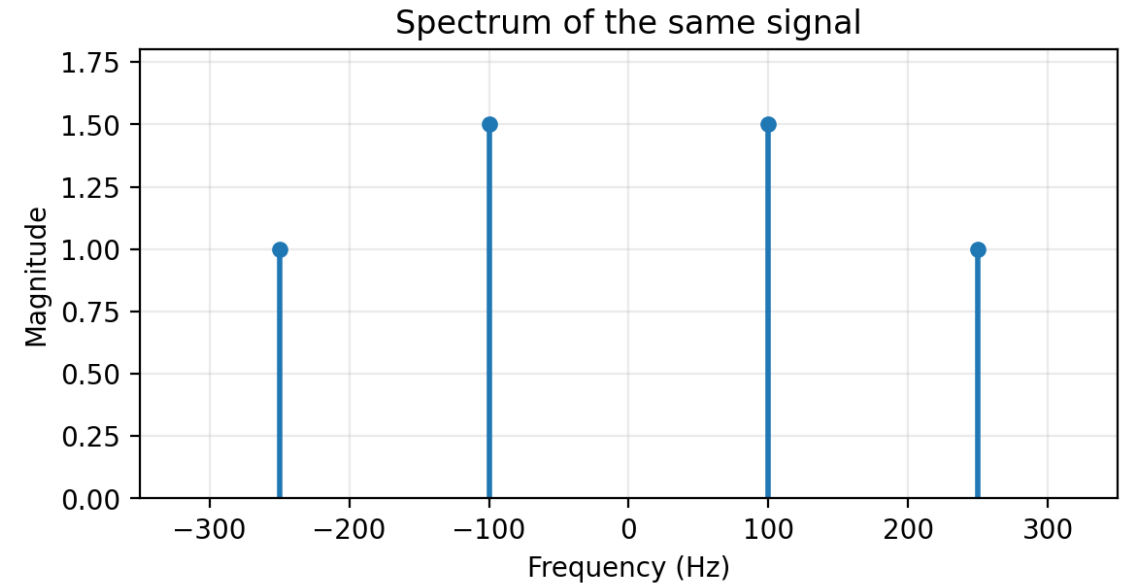
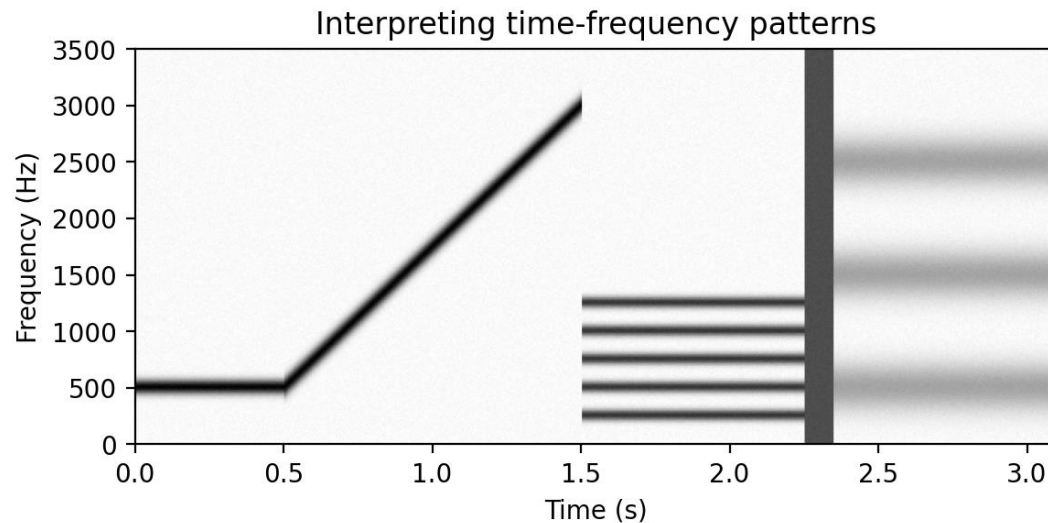


Spectrum Representation

From sums of sinusoids to Fourier series, spectrograms, and chirps

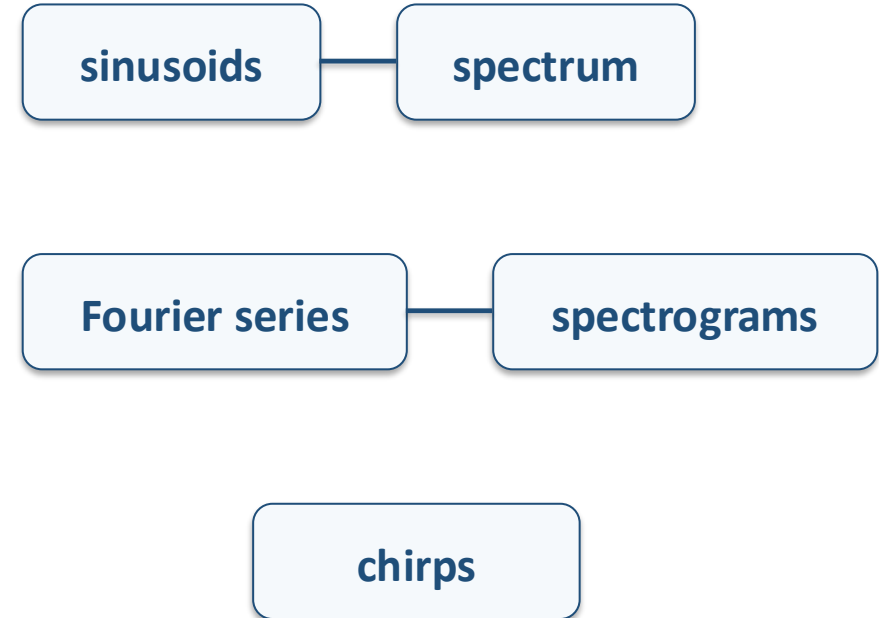


Key idea. A spectrum identifies the sinusoidal components of a signal; time-frequency representations show how those components evolve.

Learning roadmap

What students should be able to do after this topic

- Represent sums of sinusoids using two-sided spectra and complex amplitudes.
- Explain beat notes and amplitude modulation in time and frequency.
- Recognize harmonic structure and periodicity.
- Use Fourier analysis and synthesis as inverse operations.
- Interpret spectrograms and chirp signals.



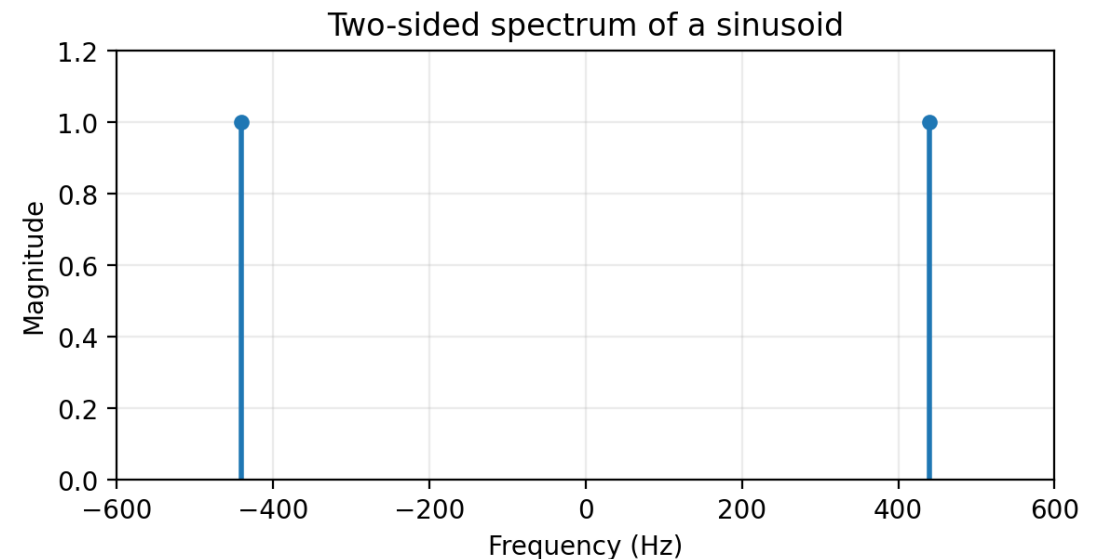
Key idea. The same signal can be studied in time, frequency, or time-frequency views.

What is a spectrum?

Frequency-domain view of sinusoidal components

$$x(t) = A \cos(2\pi f_0 t + \phi)$$

- A real cosine contains two complex rotating phasors.
- The spectrum stores frequency, magnitude, and phase.
- For real signals, positive and negative frequencies appear as conjugate partners.

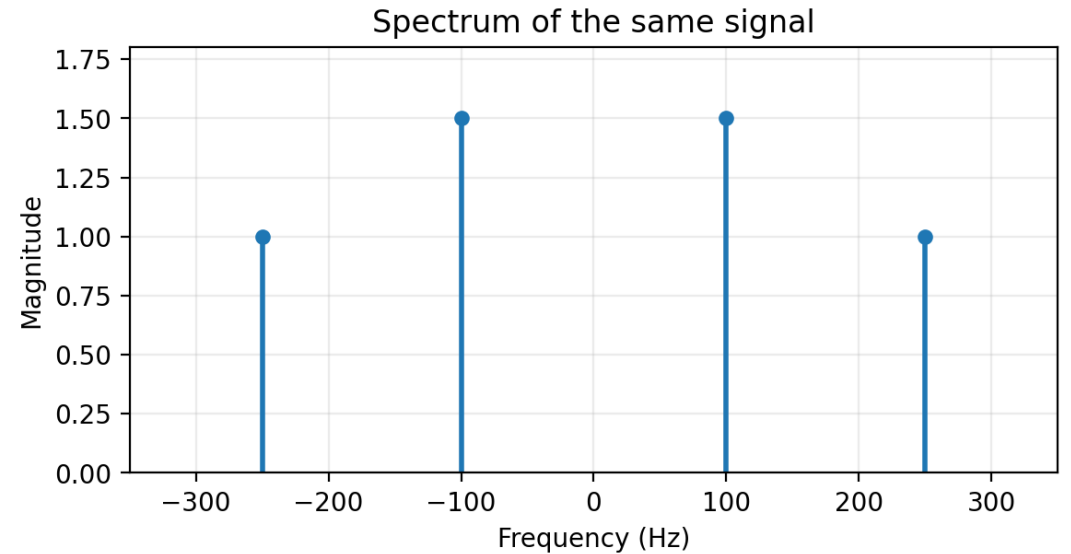
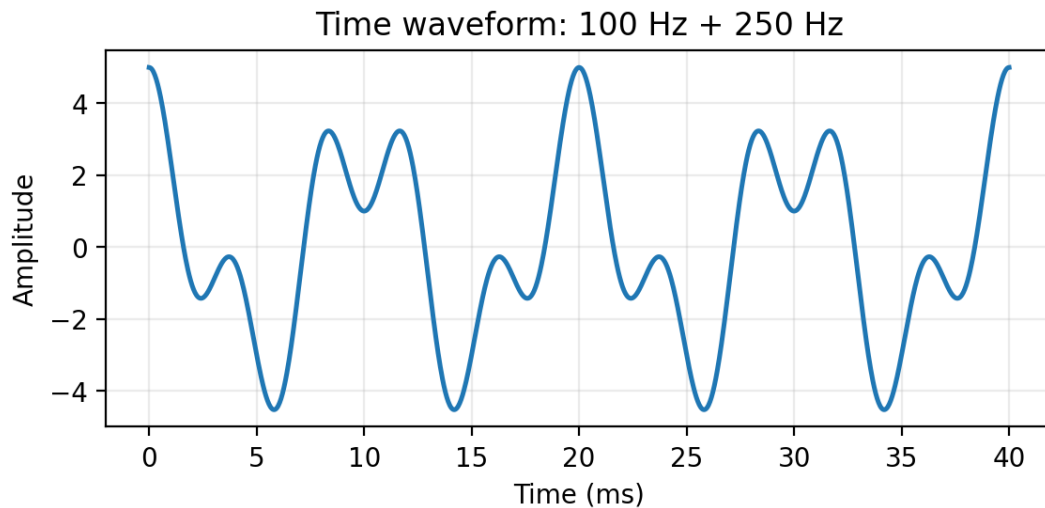


Key idea. The vertical lines show which sinusoidal frequencies are present and how strong they are.

Spectrum of a sum of sinusoids

Time waveform versus frequency-domain picture

$$x(t) = \sum_k A_k \cos(\omega_k t + \varphi_k)$$



- Linear addition builds complex waveforms.
- The spectrum is often easier to read than the waveform.
- Each sinusoid contributes a line at $\pm f_k$.

Example 1: from a sinusoid sum to a spectrum

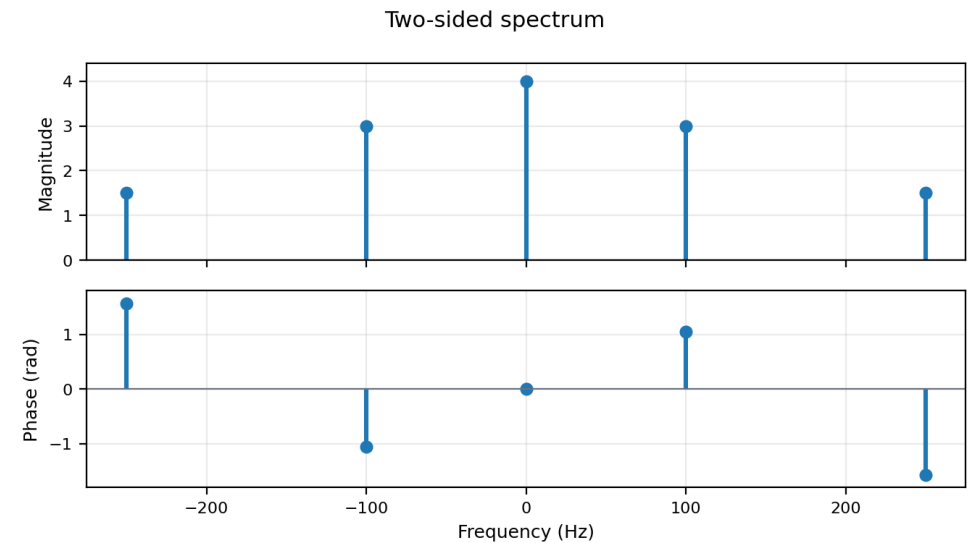
Place the DC term and each conjugate pair

$$x(t) = 4 + 6\cos(2\pi 100t + \pi/3) + 3\cos(2\pi 250t - \pi/2)$$

Question

List the two-sided spectral lines and verify conjugate symmetry.

- 1 DC line: $X(0) = 4$.
- 2 $A\cos(2\pi ft + \phi)$ gives $\frac{1}{2}Ae^{j\phi}$ at $+f$ and $\frac{1}{2}Ae^{-j\phi}$ at $-f$.
- 3 ± 100 Hz: $|X| = 3$, phases $\pm\pi/3$. ± 250 Hz: $|X| = 1.5$, phases $\mp\pi/2$.



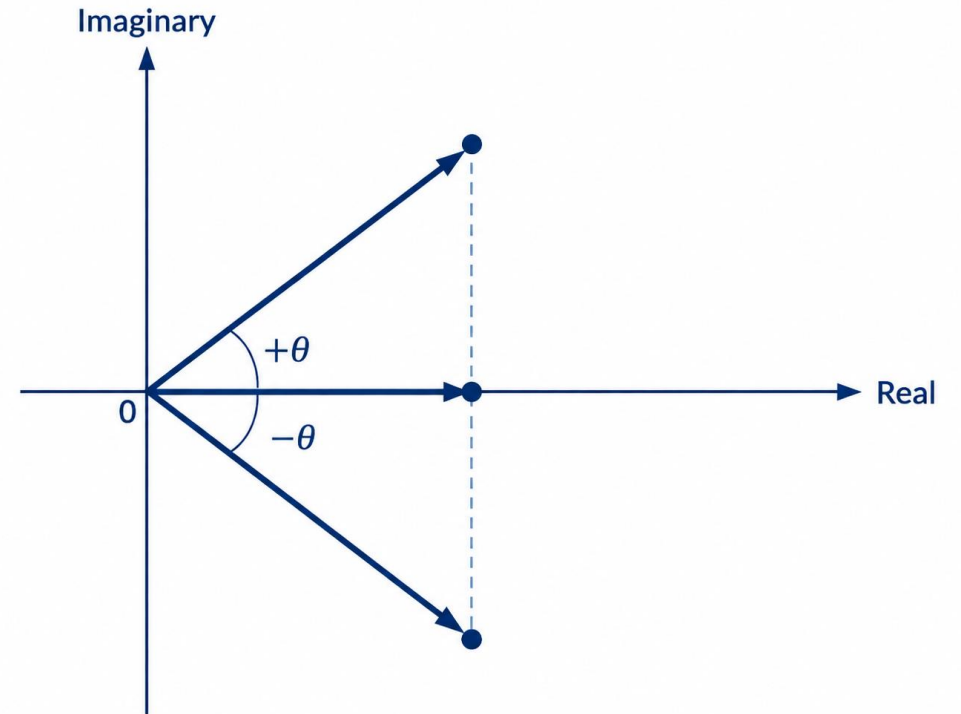
Real sinusoids as phasor pairs

Inverse Euler formula

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$A \cos(\omega_0 t + \varphi) = \frac{1}{2}Ae^{j(\omega_0 t + \varphi)} + \frac{1}{2}Ae^{-j(\omega_0 t + \varphi)}$$

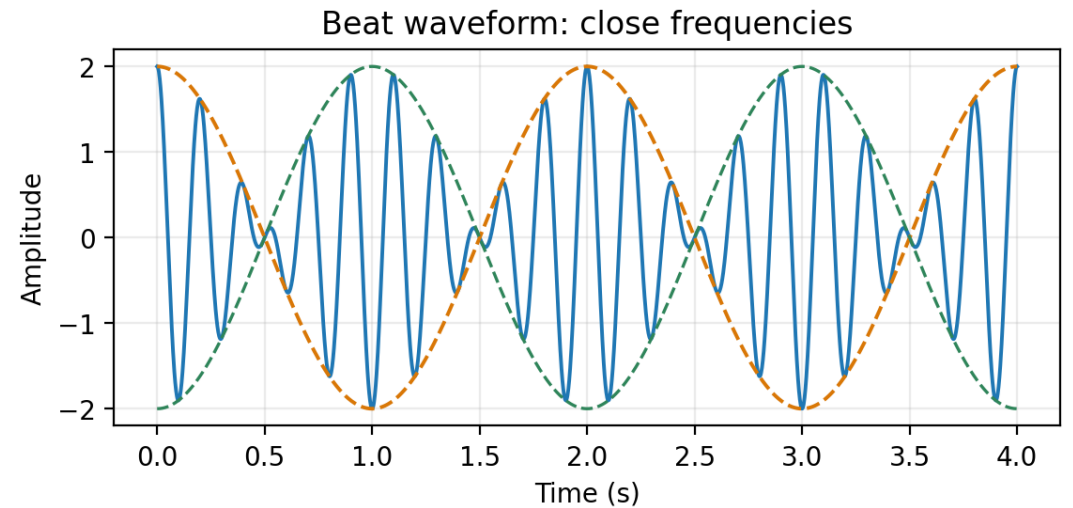
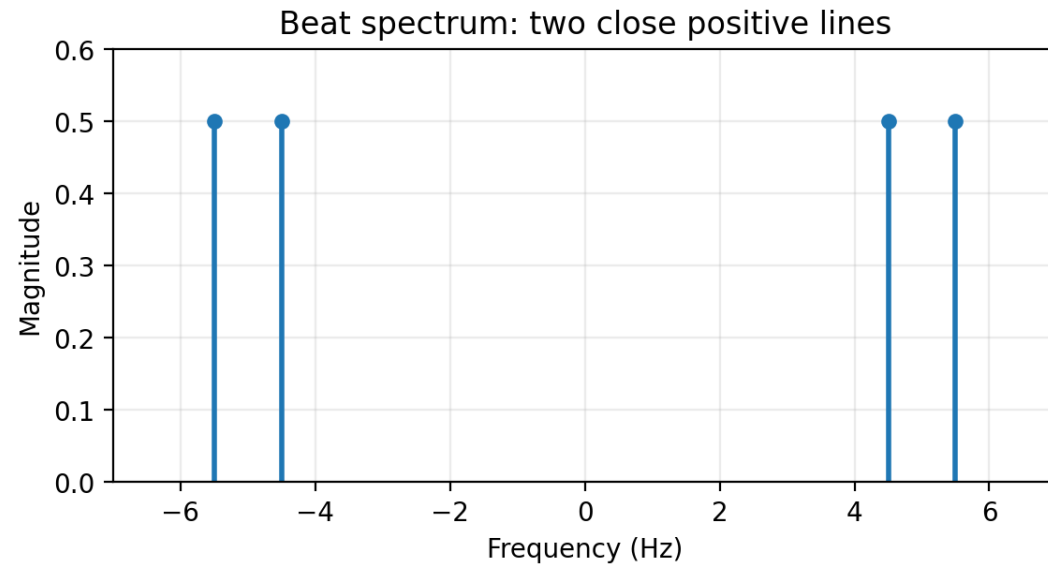
The conjugate pair makes the final signal real.



Beat notes

Close frequencies create a slow envelope

$$\cos(2\pi \cdot 4.5t) + \cos(2\pi \cdot 5.5t) = 2 \cos(2\pi \cdot 0.5t) \cos(2\pi \cdot 5t)$$



Key idea. When two frequencies are close, the waveform oscillates rapidly but its amplitude envelope varies slowly.

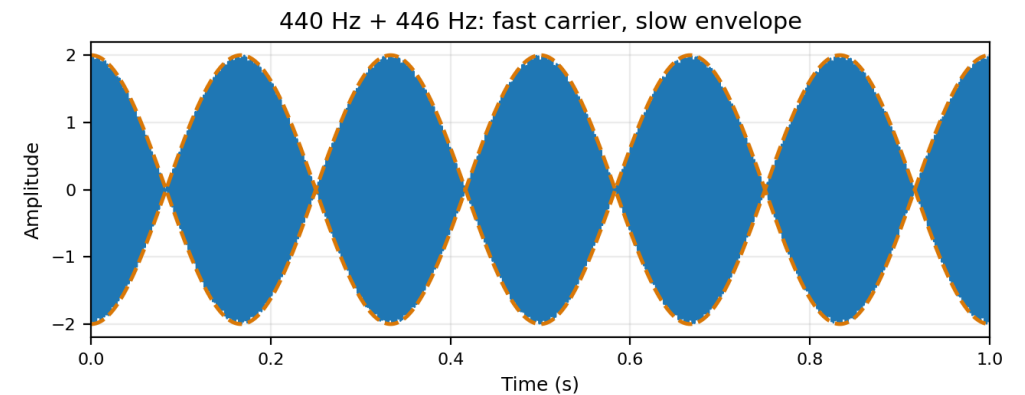
Example 2: beat notes

Close frequencies create a slow envelope

$$x(t) = \cos(2\pi 440t) + \cos(2\pi 446t)$$

- 1 Use $\cos\alpha + \cos\beta = 2\cos((\alpha-\beta)/2)\cos((\alpha+\beta)/2)$.
- 2 Centre frequency: $(440+446)/2 = 443$ Hz.
- 3 Envelope frequency: $(446-440)/2 = 3$ Hz.
- 4 Magnitude envelope has two maxima per cycle $\rightarrow$ beat rate = 6 beats/s.

$$x(t) = 2\cos(2\pi 3t)\cos(2\pi 443t)$$

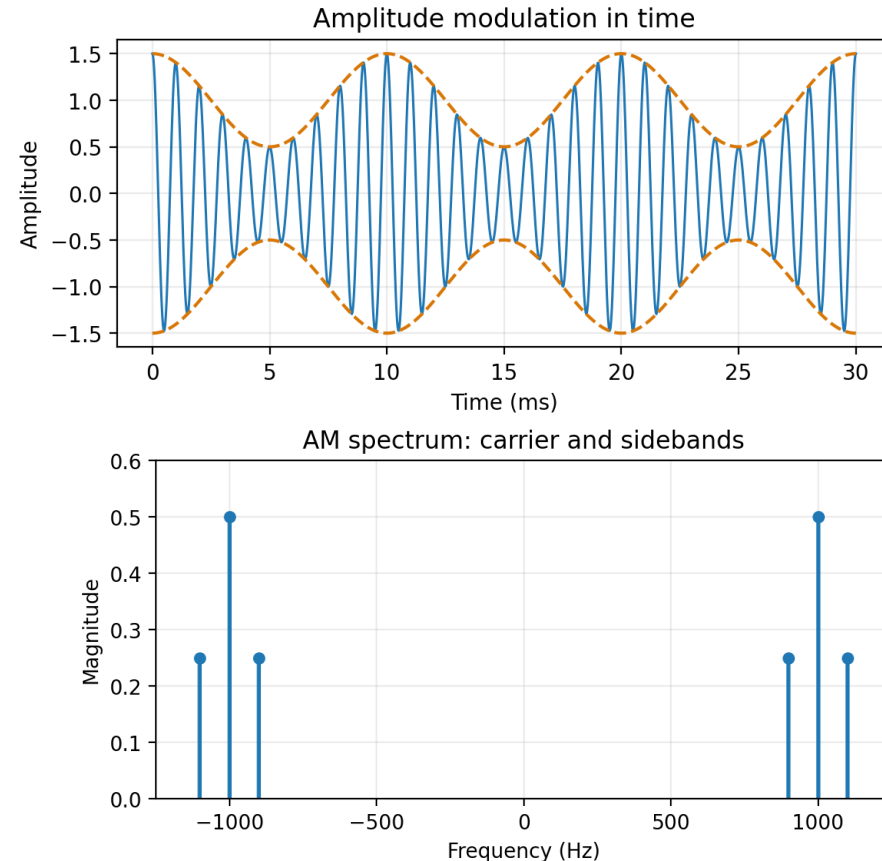


Amplitude modulation

A low-frequency signal shapes a high-frequency carrier

$$x(t) = v(t) \cos(2\pi f_c t)$$

- Carrier at f_c .
- Sidebands at $f_c \pm f_m$ for a sinusoidal modulator.
- AM envelope does not necessarily reach zero.



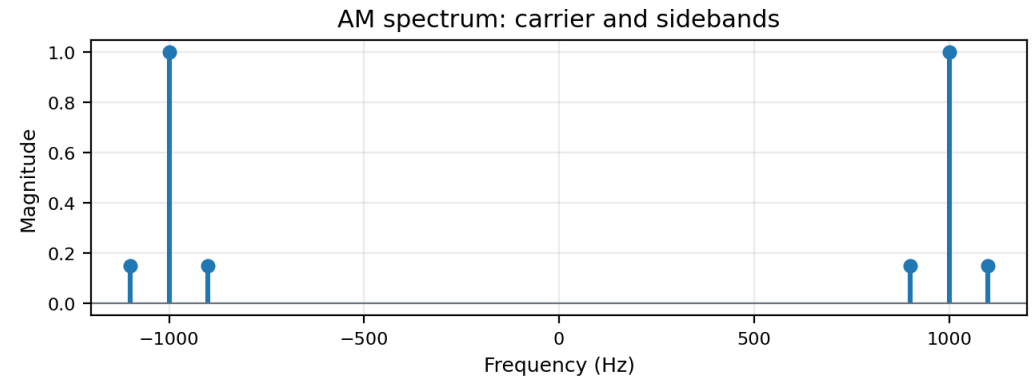
Key idea. Multiplication in time creates shifted spectral components around the carrier.

Example 3: amplitude modulation

Multiplication shifts spectra around the carrier

$$x(t) = [2 + 0.6\cos(2\pi 100t)]\cos(2\pi 1000t)$$

- 1 Distribute the carrier:
 $2\cos(2\pi 1000t) + 0.6\cos(2\pi 100t)\cos(2\pi 1000t)$.
- 2 Product-to-sum creates sidebands at 1000 ± 100 Hz.
- 3 Result: $2\cos(2\pi 1000t) + 0.3\cos(2\pi 900t) + 0.3\cos(2\pi 1100t)$.
- 4 Complex amplitudes: carrier coefficient 1; sideband coefficients 0.15.

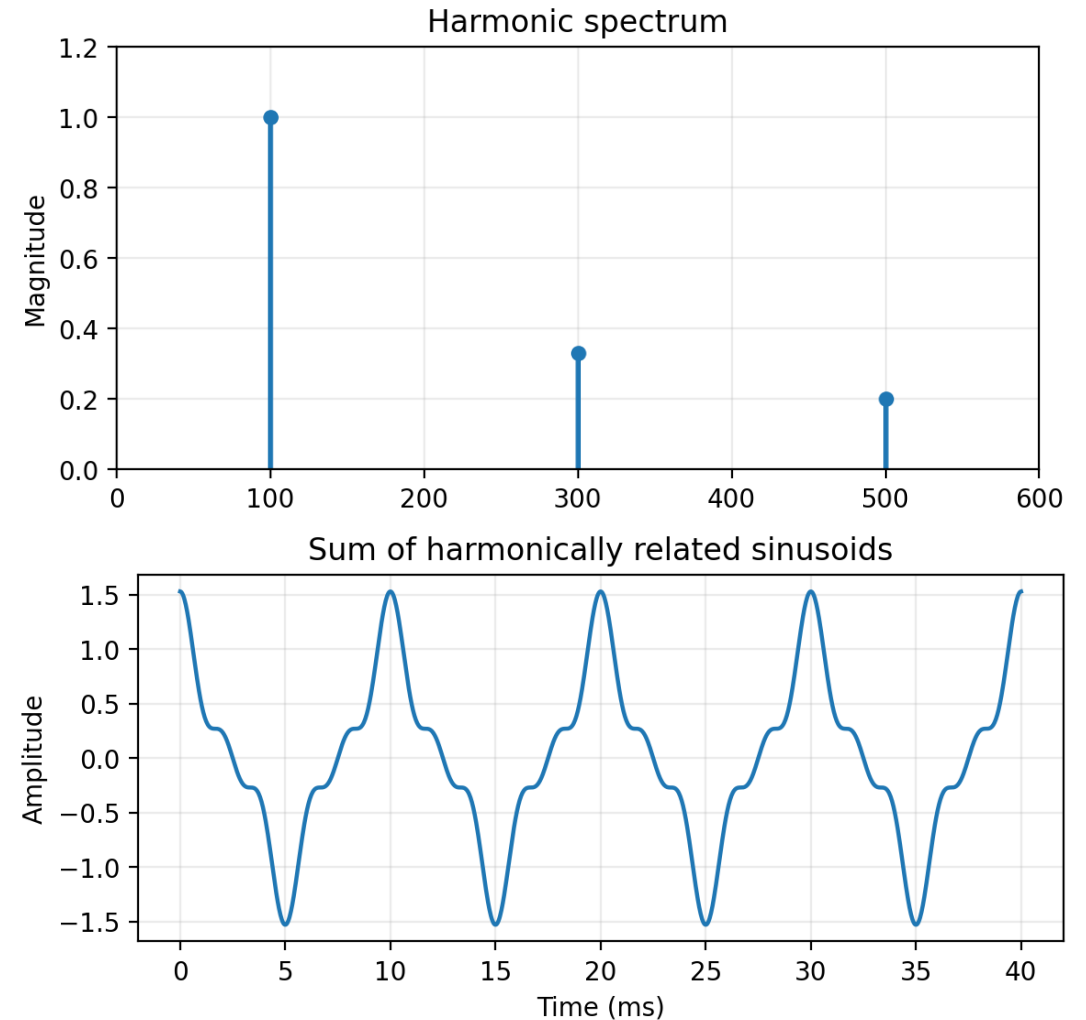


Periodic waveforms and harmonics

Integer multiples of a fundamental frequency

$$f_k = k f_0, \quad T_0 = 1 / f_0$$

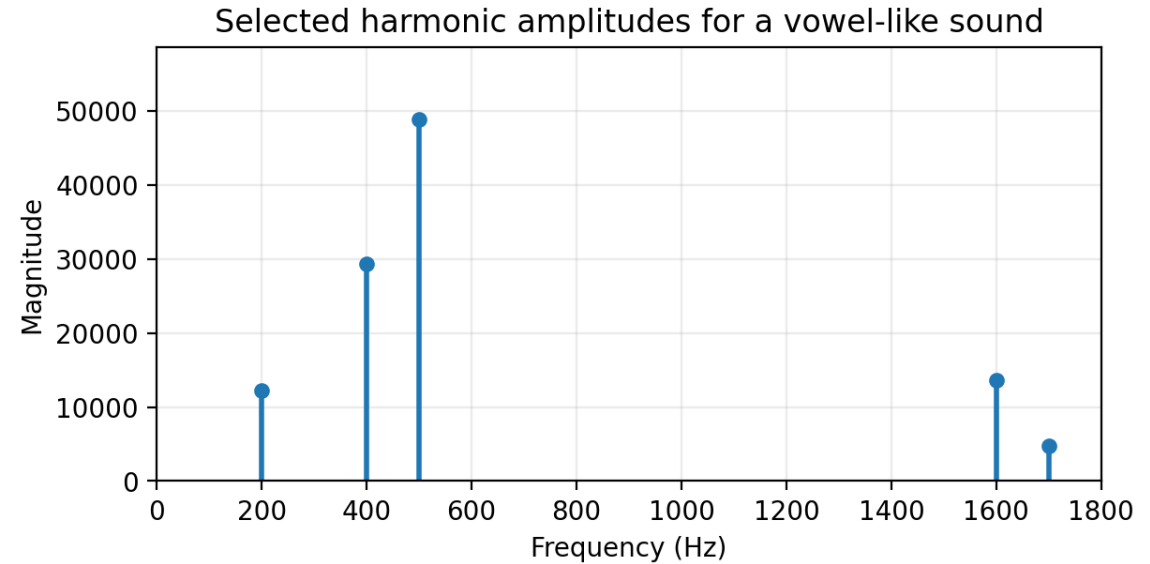
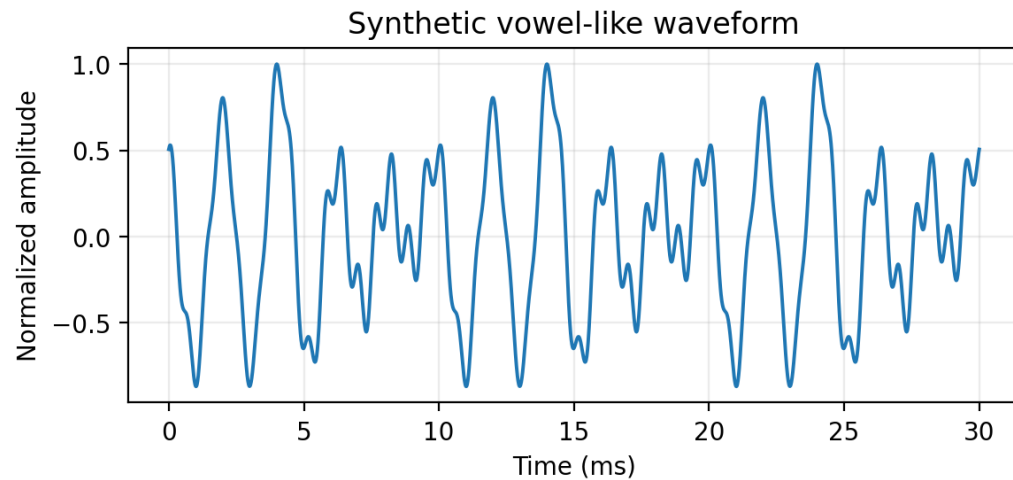
- Harmonics are frequencies at integer multiples of f_0 .
- A periodic waveform repeats every T_0 .
- Adding harmonics changes the shape but preserves periodicity.



Synthetic vowel

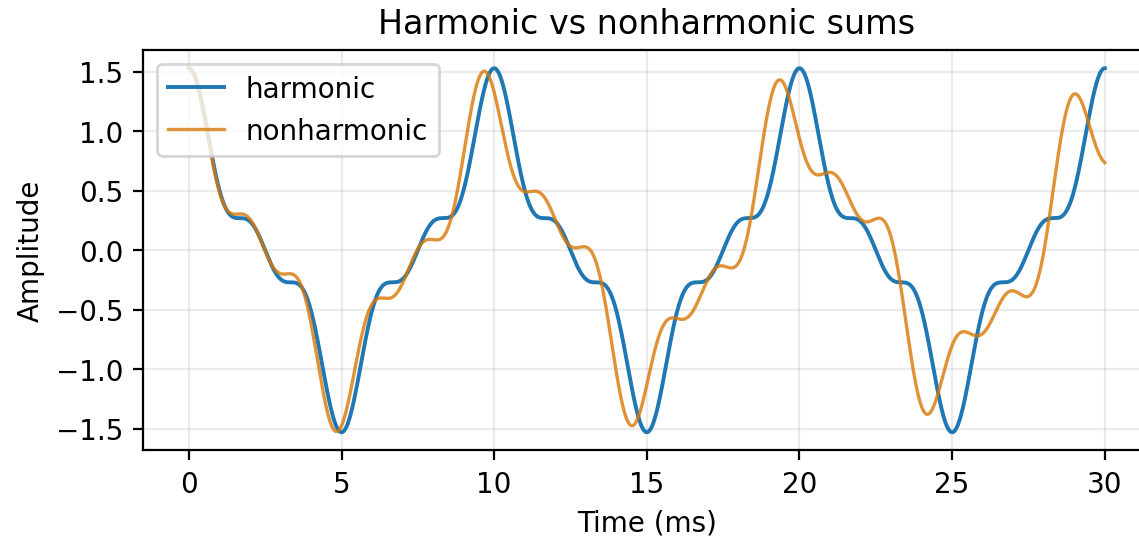
A spectrum can define a sound

$$x(t) = \sum_k X_k e^{j\omega_k t}$$



Harmonic relation and periodicity

Small frequency changes can destroy simple periodicity



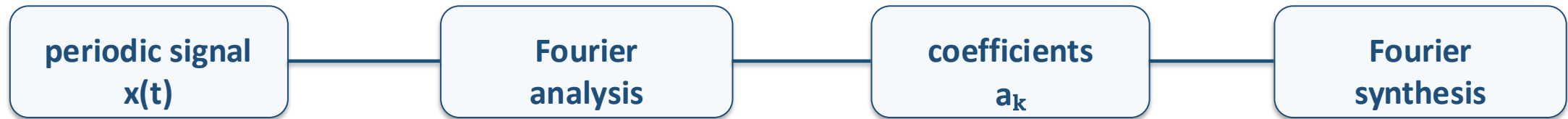
- Harmonic frequencies share a common fundamental.
- Nonharmonic frequencies may still be summed, but the result need not repeat simply.
- Spectral representation remains valid in both cases.

Key idea. Periodicity depends on frequency ratios, not on the existence of a sinusoidal sum.

Fourier series

Analysis and synthesis are complementary

$$x(t) = \sum_k a_k e^{jk\omega_0 t}$$



- Analysis: calculate spectral coefficients from a waveform.
- Synthesis: reconstruct a waveform from coefficients.
- Using finite coefficients gives an approximation.

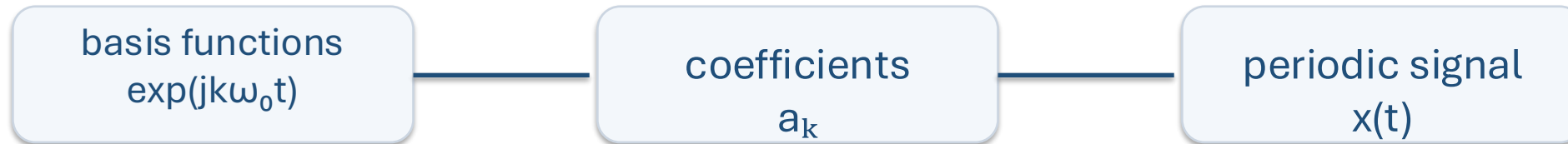
Fourier-series structure

A periodic signal is built from harmonically related exponentials

$$\omega_0 = 2\pi/T_0, \quad f_0 = 1/T_0$$

$$\omega_k = k\omega_0, \quad f_k = kf_0$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$



- Each integer k selects one harmonic frequency.
- For real signals, negative-frequency coefficients are conjugates: $a_{-k} = a_k^*$.

Why Fourier analysis works

Orthogonality of complex exponentials

$$a_k = (1/T_0) \int_0^{T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\int_0^{T_0} e^{j(k-m)\omega_0 t} dt = 0 \quad \text{for } k \neq m$$

- Basis functions repeat over the same period.
- Integrating against one basis function isolates its coefficient.
- The DC coefficient a_0 is the average value.

Key idea. Fourier analysis is a projection onto harmonically related complex exponentials.

Deriving the Fourier-analysis equation

Multiply by one basis function and integrate over a period

Assume: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$

Multiply by $e^{-jm\omega_0 t}$ and integrate

$$\int_0^{T_0} x(t) e^{-jm\omega_0 t} dt = \sum_{k=-\infty}^{\infty} a_k \int_0^{T_0} e^{j(k-m)\omega_0 t} dt$$

$$\int_0^{T_0} e^{j(k-m)\omega_0 t} dt = 0 \quad (k \neq m), \quad = T_0 \quad (k = m)$$

Therefore: $\int_0^{T_0} x(t) e^{-jm\omega_0 t} dt = T_0 a_m$

$$a_m = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jm\omega_0 t} dt$$

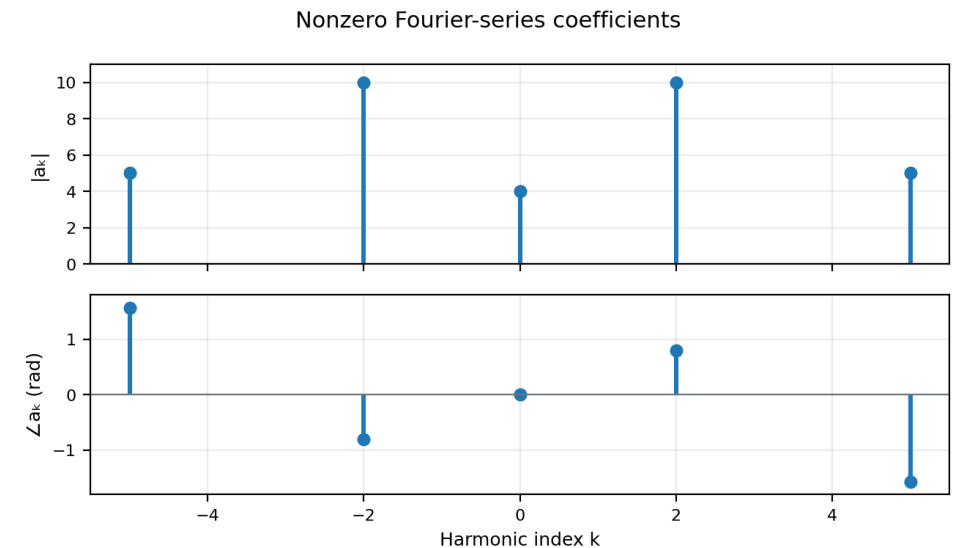
Interpretation: Fourier analysis is a projection onto one harmonic basis function.

Example 4: Fourier-series coefficients

Find the common fundamental and place the coefficients

$$x(t) = 4 + 20\cos(2\pi 100t + 0.8) + 10\cos(2\pi 250t - \pi/2)$$

- 1 100 Hz and 250 Hz share $f_0 = \gcd(100, 250) = 50$ Hz.
- 2 100 Hz = $2f_0 \rightarrow$ coefficients at $k=\pm 2$ with magnitudes 10.
- 3 250 Hz = $5f_0 \rightarrow$ coefficients at $k=\pm 5$ with magnitudes 5.
- 4 Phases: a_2 has $+0.8$, a_{-2} has -0.8 ; a_5 has $-\pi/2$, a_{-5} has $+\pi/2$.

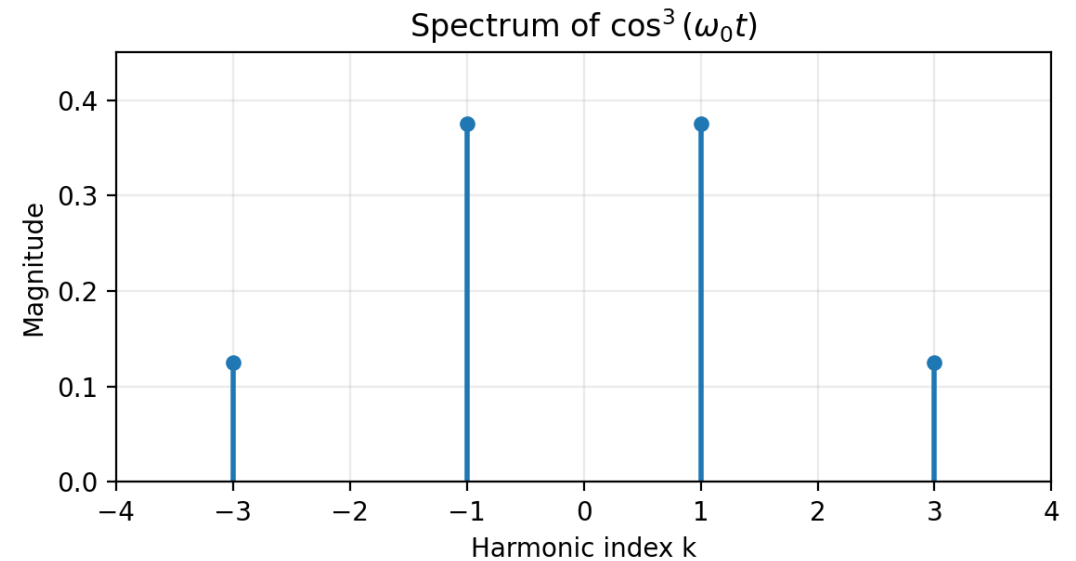


Spectrum from Fourier-series coefficients

Example: a cubed sinusoid

$$\cos^3(\theta) = (3/4)\cos(\theta) + (1/4)\cos(3\theta)$$

- Only harmonics $k = \pm 1$ and $k = \pm 3$ are present.
- The third harmonic appears because of the nonlinear operation.
- The two-sided spectrum shows positive and negative harmonic indices.



Use a trigonometric identity to read the Fourier coefficients

$$x(t) = \cos^3(\omega_0 t)$$

$$\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos 3\theta$$

- 1 The signal contains only the fundamental and the third harmonic.

$$x(t) = \frac{3}{4} \cos(\omega_0 t) + \frac{1}{4} \cos(3\omega_0 t)$$

- 2 A real cosine of amplitude A gives Fourier coefficients A/2 at $k=+r$ and $k=-r$.

$$a_1 = a_{-1} = \frac{3}{8}, \quad a_3 = a_{-3} = \frac{1}{8}, \quad \text{all other } a_k = 0$$



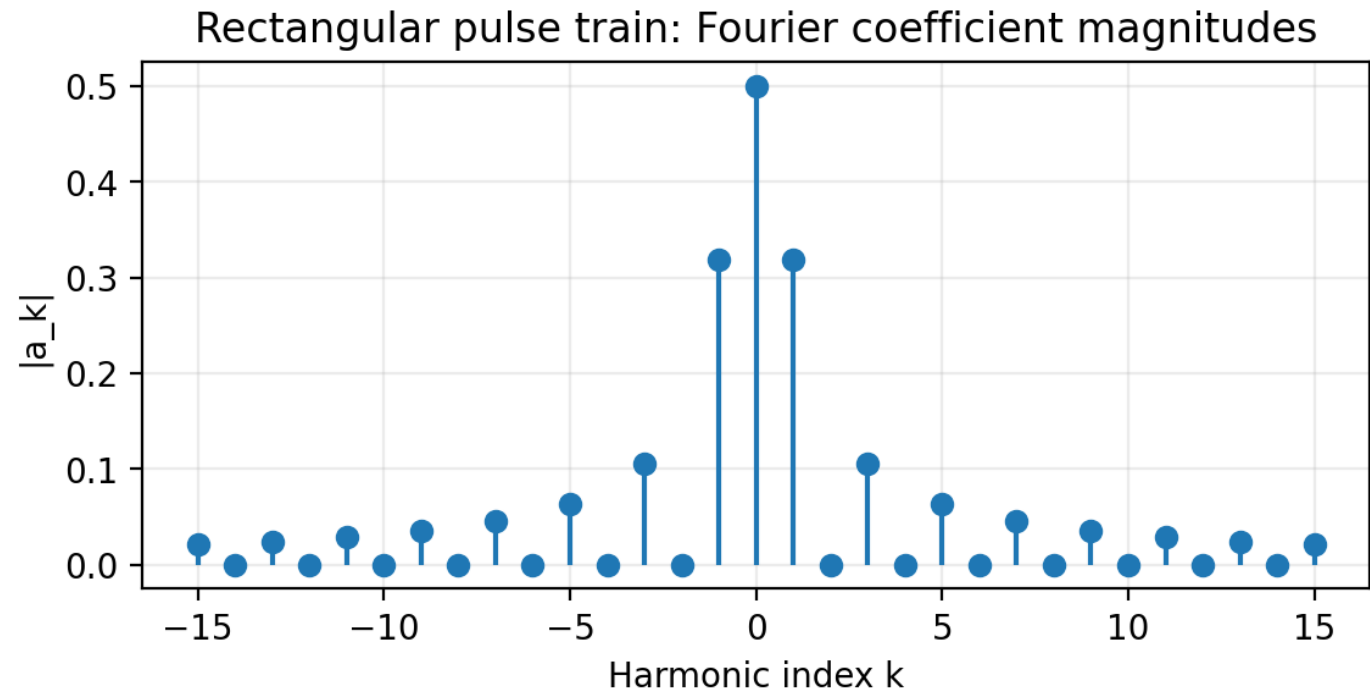
Fourier analysis of periodic signals

Rectangular pulse train



rectangular pulse train

- Sharp edges require many harmonics.
- For a 50% duty cycle, only odd harmonics are present.
- Coefficient magnitudes decay roughly as $1/|k|$.



Derivation: rectangular pulse-train coefficients

Integrate only over the interval where the pulse is nonzero

$$x(t) = 1 \text{ for } |t| \leq \tau/2, \quad x(t) = 0 \text{ otherwise,} \quad T_0 \text{ periodic}$$

$$a_k = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} e^{-jk\omega_0 t} dt$$

$$a_k = \frac{1}{T_0} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-\tau/2}^{\tau/2}$$

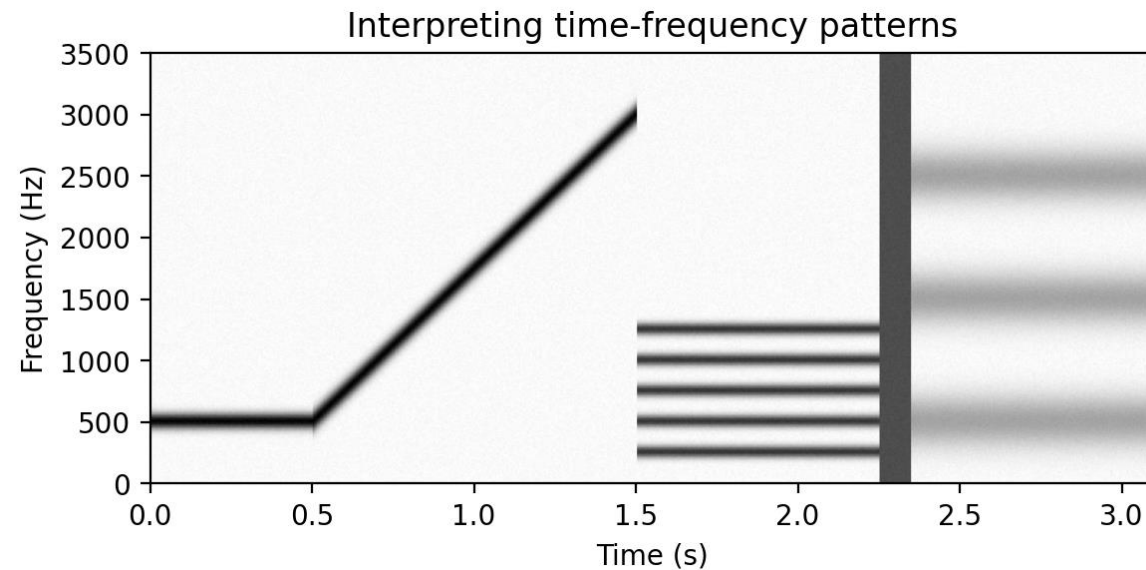
$$a_k = \frac{\sin(k\omega_0 \tau/2)}{k\pi}, \quad k \neq 0$$

$$a_0 = \frac{\tau}{T_0}, \quad a_k = \frac{\tau}{T_0} \operatorname{sinc}\left(\frac{k\tau}{T_0}\right)$$

- For 50% duty cycle, $\tau/T_0=1/2$: even harmonics vanish and odd harmonics decay roughly as $1/|k|$.
- Slow coefficient decay explains why many harmonics are needed to reconstruct sharp edges.

Time-frequency spectrum

When spectral content changes over time



- Horizontal lines: stationary tones.
- Diagonal ridges: changing frequency.
- Stacks: harmonically related partials.
- Vertical structures: short broadband events.

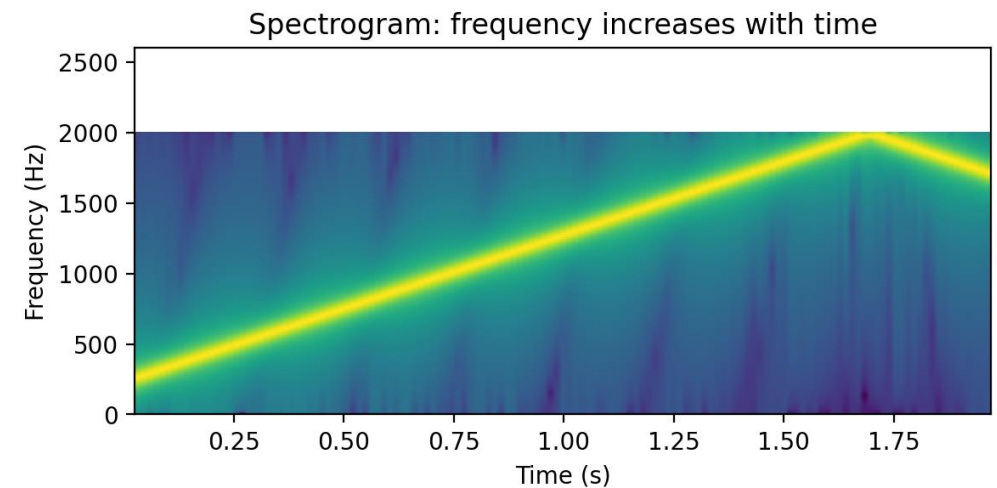
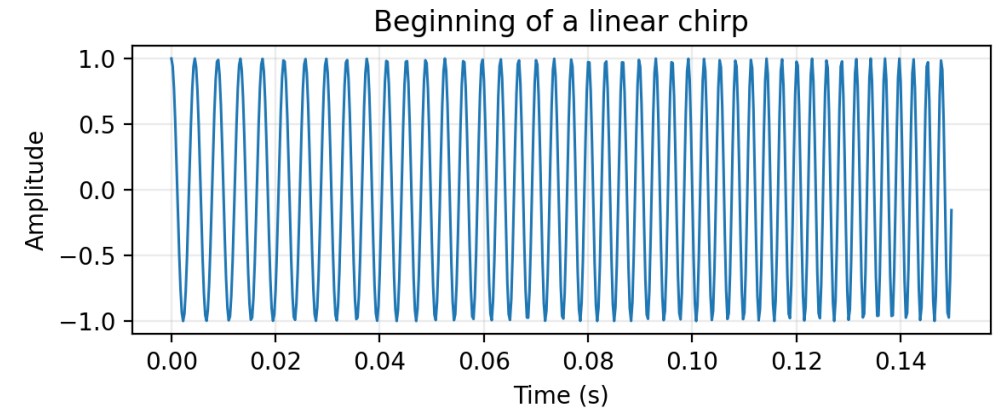
Key idea. A spectrogram adds a time axis to spectral analysis.

Frequency modulation and chirps

Changing phase creates changing frequency

$$\omega_i(t) = d\Phi(t)/dt, \quad f_i(t) = \omega_i(t)/(2\pi)$$

- A chirp has continuously changing frequency.
- A quadratic phase produces linearly changing instantaneous frequency.
- In a spectrogram, the chirp appears as a sloped ridge.

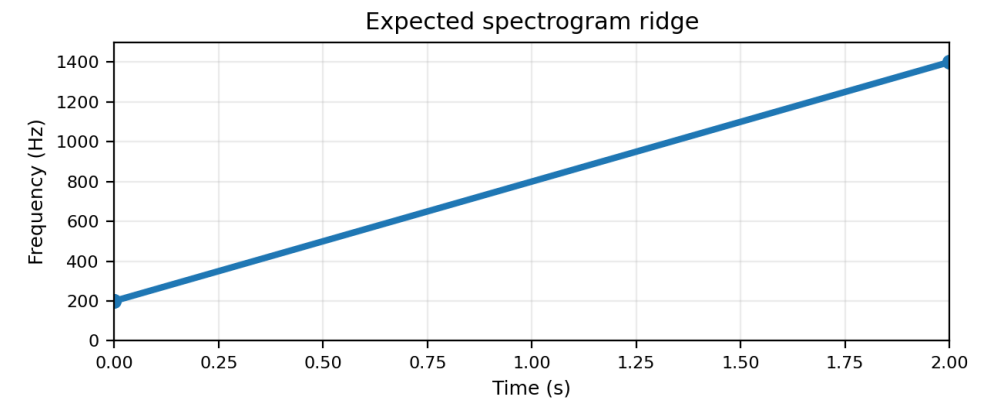


Example 5: instantaneous frequency of a chirp

Differentiate phase to read the spectrogram ridge

$$x(t) = \cos(2\pi(200t + 300t^2)), \quad 0 \leq t \leq 2 \text{ s}$$

- 1 Phase: $\Phi(t) = 2\pi(200t + 300t^2)$.
- 2 Instantaneous radian frequency: $\omega_i(t) = d\Phi/dt = 2\pi(200 + 600t)$.
- 3 Instantaneous frequency: $f_i(t) = \omega_i(t)/(2\pi) = 200 + 600t \text{ Hz}$.
- 4 From 200 Hz at $t=0$ to 1400 Hz at $t=2 \text{ s}$: a rising straight ridge.



Notation and takeaways

Minimum language for spectrum representation

$x(t)$	continuous-time signal
X_k	complex amplitude / phasor
ω_k, f_k	radian and cyclic frequency
ω_o, f_o	fundamental frequencies
a_k	Fourier-series coefficient
$\Phi(t), \omega_i(t)$	time-varying phase and instantaneous frequency

- A spectrum is a compact frequency-domain description.
- Fourier series gives the spectrum of periodic signals.
- Spectrograms and chirps extend the idea to time-varying signals.